11 **(a)** Fig. 11.1 shows the paths of three α-particles moving towards a thin gold foil. Four gold nuclei are shown.

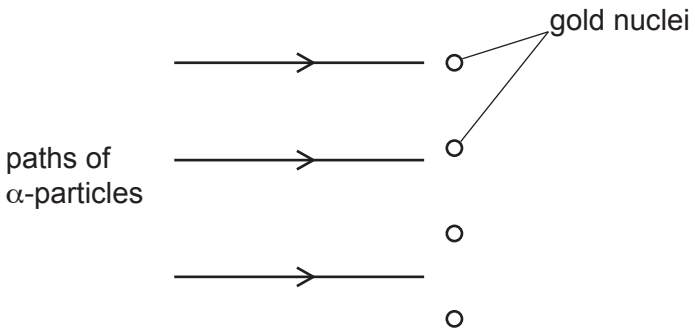


Fig. 11.1 (not to scale)

(i) On Fig. 11.1, complete the paths of the **three** α-particles. [3]

(ii) State the sign of the charge on the α-particles.

..... [1]

(b) The nuclide notation for a nucleus of gold-198 is $^{198}_{79}\mathrm{Au}$.

State the numbers of electrons, neutrons and protons in a neutral atom of gold-198.

number of electrons = ...

number of neutrons = ...

number of protons = ...

[3]

[Total: 7]

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